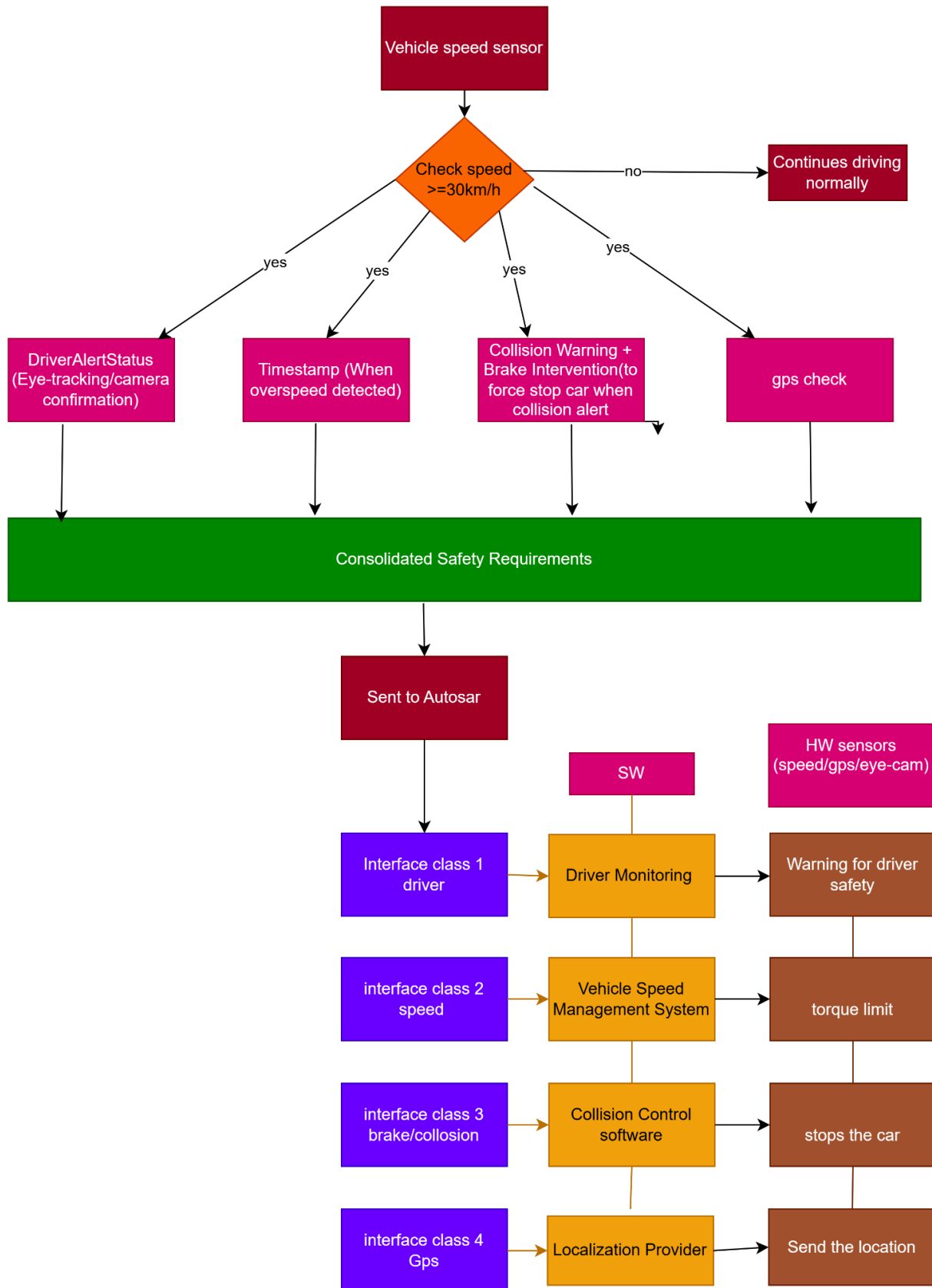




1. Driver Monitoring Requirements (Interface Class 1)

- **SRS:** The system shall detect driver distraction or drowsiness via eye-tracking data when vehicle speed exceeds 30km/h
- **SWS:** The **Driver_Monitor_SWC** shall receive **EyeTracking_Signal** via the RTE and trigger an **HMI_Warning_Signal** if pupils are not detected for >2 seconds.

2. Speed & Engine Management (Interface Class 2)

This maps the "Check speed >= 30km/h" logic to the "Torque Limit" output.

- **SRS:** The vehicle shall automatically limit torque if the current speed exceeds the limit.
- **SWS:** The **Engine_Management** shall calculate a **Torque_Limit_Request** based on the data between **Vhc_Speed_Sensor** and **GPS_Speed_Limit** signals.

3. Collision & Brake Intervention (Interface Class 3)

This maps the "Brake Intervention" logic to the "Stops the car" output.

- **SRS:** The system shall initiate emergency braking if an obstacle is detected while the vehicle is in an overspeed state.
- **SWS:** The **Collision control** shall send a high-priority **Brake_Pressure_Cmd** to the **Brake Actuator** if the **Collision_Risk_Level** exceeds threshold

4. Localization & Timestamp (Interface Class 4)

This handles the "GPS check" and "Timestamp"

- **SRS:** Every overspeed event must be logged with a synchronized UTC timestamp
- **SWS:** It should send the exact longitude and latitude